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Department of Computer Science and Engineering

Academic Year 2025 – 2026 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. CSE

Course Code & Title: CS3452 – Theory of Computation

Name of the Faculty Member(s): Mrs. S. Manjula

Innovative Practice Description

Unit / Topic: Unit III / CFG to PDA – Context Free Grammar to Pushdown Automaton

Course Outcome: CO 3

Topic Learning Outcome: TLO 9

Activity Chosen: Pioneers to Peers

Date of Activity: 18.02.2026

Pioneer Student: ARUNADEVI M (953623104012), III CSE

Justification:

- The "Pioneer to Peer" initiative promotes collaborative learning by allowing advanced students to share knowledge with their juniors. The topic "CFG to PDA – Converting Context Free Grammar to Pushdown Automaton" was selected as it is a foundational concept in Theory of Computation, often perceived as challenging. Peer explanation in simple, relatable terms helps students grasp abstract computational concepts more effectively.
- CFG to PDA conversion is a core topic in Theory of Computation that involves abstract reasoning about formal languages and automata.
- Peer-led explanation helps demystify complex theoretical steps, making concepts more approachable for learners.
- The "Pioneer to Peer" approach builds communication skills for the senior student and provides relatable mentorship for juniors.
- Step-by-step board demonstration reinforces both the teacher's and students' understanding of systematic conversion procedures.

Time Allotted for the Activity: 35 Minutes

Details of the Implementation:

The activity was conducted during a scheduled class hour with II Semester B.E. CSE students as part of the Pioneer to Peer initiative. The focus of the session was on "CFG to PDA – Converting Context Free Grammar to Pushdown Automaton", a core topic that introduces students to the construction of abstract machines from formal grammars. The implementation involved the following steps:

Step 1: Introduction to CFG and PDA

- ARUNADEVI M (953623104012), III CSE began the session by introducing the concept of Context Free Grammar (CFG) and Pushdown Automata (PDA) and their relationship in formal language theory.
- She briefly recapped CFG components: variables, terminals, production rules, and start symbol.

Step 2: Explaining the Conversion Process

- She explained the systematic steps to convert a given CFG into an equivalent PDA.
- The key rules for conversion were clearly written on the whiteboard: for each production rule $A \rightarrow \alpha$, a transition is added to the PDA.
- She demonstrated how the stack of the PDA models the derivation steps of the CFG.

Step 3: Step-by-Step Problem Solving

- A sample CFG was taken: $S \rightarrow aSb \mid \epsilon$ (generating the language $a^n b^n$).
- ARUNADEVI M traced through the conversion step by step, writing each PDA transition on the board.
- She illustrated stack operations (push/pop) corresponding to each production rule during string derivation.
- Halting conditions and acceptance by empty stack were clearly explained.

Step 4: Interactive Learning

- Students were given additional CFG examples to convert to PDA on their own.
- ARUNADEVI M monitored student attempts, provided hints, and corrected misconceptions in real time.
- Students actively participated and cross-checked their answers with the solutions on the board.

CO – PO / PSO Mapping:

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO 3	2	2	2	1	1	1

1–Low 2–Moderate 3–High

PO / PSO Mapped:

Innovative Practice	PO1	PO2	PO3	PO9	PO10	PSO1
Justification for correlation	2	2	2	1	1	1

PO / PSO Justification:

PO / PSO	Justification
PO1	Students applied mathematics, formal language theory, and engineering knowledge to convert CFG into PDA and solve computational problems.
PO2	Students analyzed the problem of CFG to PDA conversion and verified the solution using formal language concepts.
PO3	Students designed a Pushdown Automaton that satisfied the requirements of the given Context Free Grammar.
PO9	Students worked effectively as a team during the peer-led activity by discussing and solving CFG to PDA conversion problems together.
PO10	Students improved their communication skills by understanding and explaining the step-by-step CFG to PDA conversion process.
PSO1	Students will identify the problem and successfully convert CFG into PDA using theoretical computer science concepts

Images / Screenshot of the Practice:

Fig 1: ARUNADEVI M writing CFG to PDA conversion steps on the board



Fig 2: Session overview – students attending the Pioneer to Peer lecture



Fig 3: ARUNADEVI M explaining PDA transition rules interactively



Fig 4: Students engaged and listening attentively during the session

Reflective Critique:

Feedback of Practice from Students and Other Stakeholders:

- Students responded positively and appreciated the approachable and relatable teaching style of their senior ARUNADEVI M.
- The step-by-step CFG to PDA conversion explanation helped students understand the logical structure of the process clearly.
- Students felt more confident in approaching Theory of Computation problems after the session.

Benefit of the Practice:

- Enhanced interaction and bonding between junior and senior students.
- Encouraged knowledge sharing and boosted communication skills of the pioneer student.
- Improved understanding of abstract theoretical concepts through peer-level explanation.
- Most students actively participated and successfully performed CFG to PDA conversions during the session.

Challenges Faced in Implementation:

- A small number of students were initially hesitant to participate actively.
- These students were encouraged by highlighting the personal benefits of self-learning and peer engagement.

References:

1. <https://www.ishcmc.com/news-and-blog/peer-led-learning/>
2. <https://www.cascadiaschool.org/what-are-the-benefits-of-mixing-ages-and-grades-in-a-classroom-doesnt-this-make-teaching-way-harder-sorryteachers>
3. <https://www.21kschool.com/in/blog/benefits-of-activity-based-learning/>

Signature of Faculty Member

HOD / CSE